



Mixed-Type Patient Clustering Benchmark

A CSV-first comparison of eight mixed-data clustering strategies

60 patients | 12 mixed features | 6 clusters fitted

The report analyzes `example_patients_input.csv` using distance-based, prototype-based, dimension-reduced, probabilistic, spectral, and consensus clustering. Dedicated heatmaps show the patient structure produced by every method.

Reproducible seed: 20260902

Executive summary

Headline results from the executed benchmark

6.2%

Overall missing cells

0.255

Consensus silhouette

68.2%

Mean membership certainty

- No truth labels were supplied. FAMD-style + k-means had the strongest subsample stability (0.829); the grand consensus stability was 0.553.
- Across MAR-style imputations, the consensus partitions had mean pairwise ARI 0.600. This isolates uncertainty associated with missing values from ordinary resampling instability.
- Directional MNAR scenarios reassigned between 0.0% and 0.0% of patients relative to the baseline scenario.
- No single internal index proves that 6 biological subtypes exist. Retain K=1 as a possibility and require clinical interpretability, resampling stability, and external replication.
- No external recovery metric is shown because no truth column was supplied. Internal and stability metrics remain available, but external validation is still needed.

Input data overview

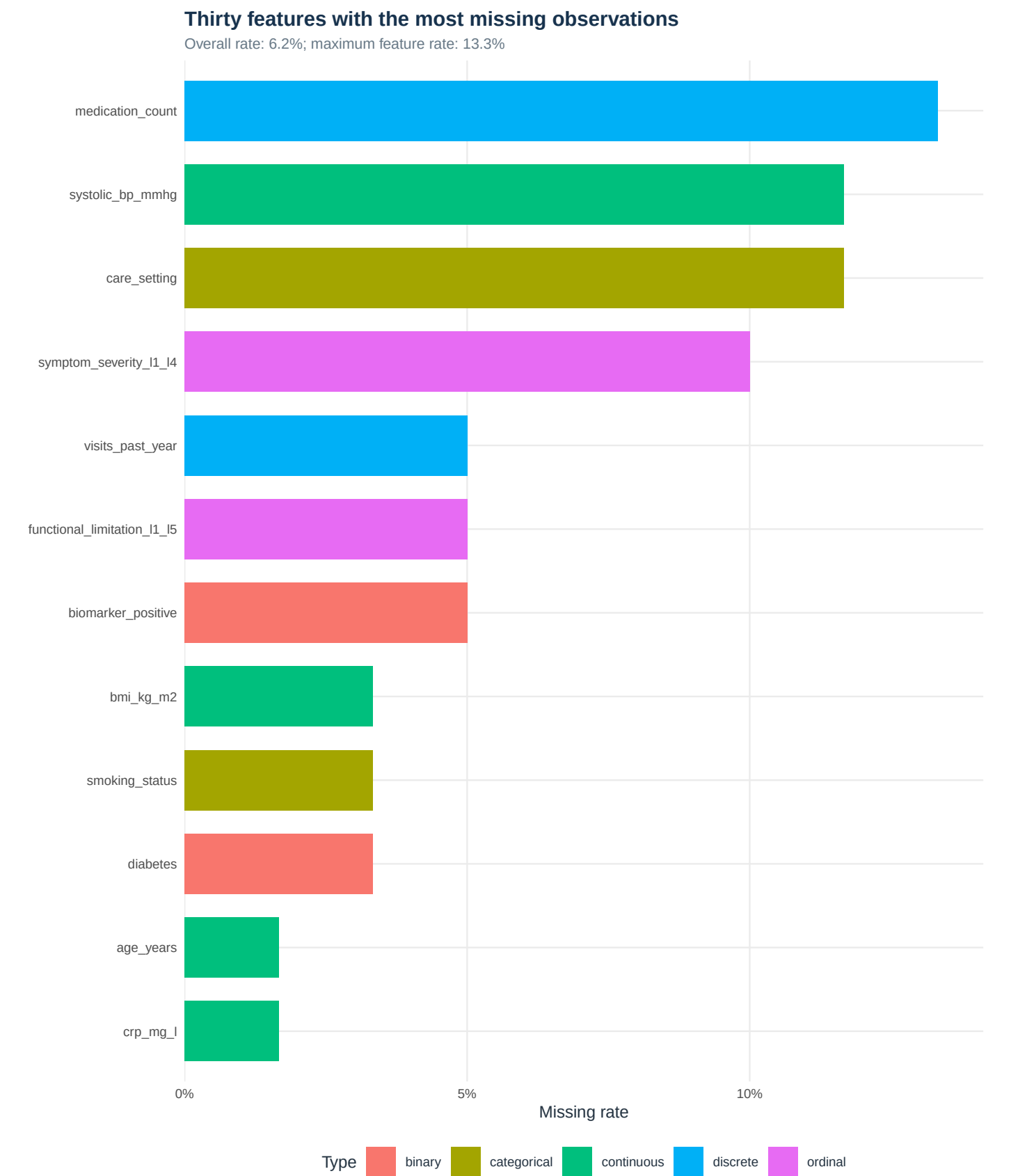
The feature dictionary controls types, domains, and excluded metadata

Feature type	Features	Mean missing
continuous	4	4.6%
discrete	2	9.2%
ordinal	2	7.5%
categorical	2	7.5%
binary	2	4.2%

- Input: example_patients_input.csv. The analysis used 60 rows and 12 clustering features.
- Feature types were inferred from the CSV and saved to feature_dictionary_used.csv. Review ordinal variables and clinical domains before treating this as a final analysis.
- The dictionary contains 12 weighting domains. Each domain receives equal total Gower weight, so large feature blocks do not dominate solely by column count.
- Observed missingness is 6.2% overall. Stochastic mixed-type hot-deck imputations preserve observed categories and propagate assignment uncertainty.
- No truth column was supplied; ARI, NMI, and label-matched accuracy are therefore intentionally unavailable.

Observed missingness

Feature-level rates; colors indicate declared measurement type



Approaches implemented

All methods use the same type dictionary and imputed datasets

Approach	Representation	Assignment	Key caveat
Gower + PAM	Weighted mixed dissimilarity	Medoid	Feature weighting
Gower + hierarchical	Weighted mixed dissimilarity	Average-linkage cut	Linkage sensitivity
k-prototypes	Standardized numeric + category modes	Prototype	Lambda and local minima
FAMD-style + k-means	Balanced mixed latent axes	Centroid	Two-stage approximation
Latent-embedding mixture	Balanced mixed latent axes	Posterior mode	Latent Gaussian assumption
Latent class/profile	Typed conditional distributions	Posterior mode	Conditional independence
Gower spectral	Gower affinity eigenvectors	Spectral centroid	Kernel bandwidth
Grand consensus	Co-assignment matrix	Consensus medoid	Can average weak answers

Implementation note: the FAMD-style embedding is a dependency-light PCAmix approximation. It standardizes numeric/ordinal variables and balances each nominal indicator block before PCA. Production work can replace it with FactoMineR::FAMD or PCAmixdata without changing the evaluation harness.

Method performance

Consensus per method across eight imputations; higher is better except runtime

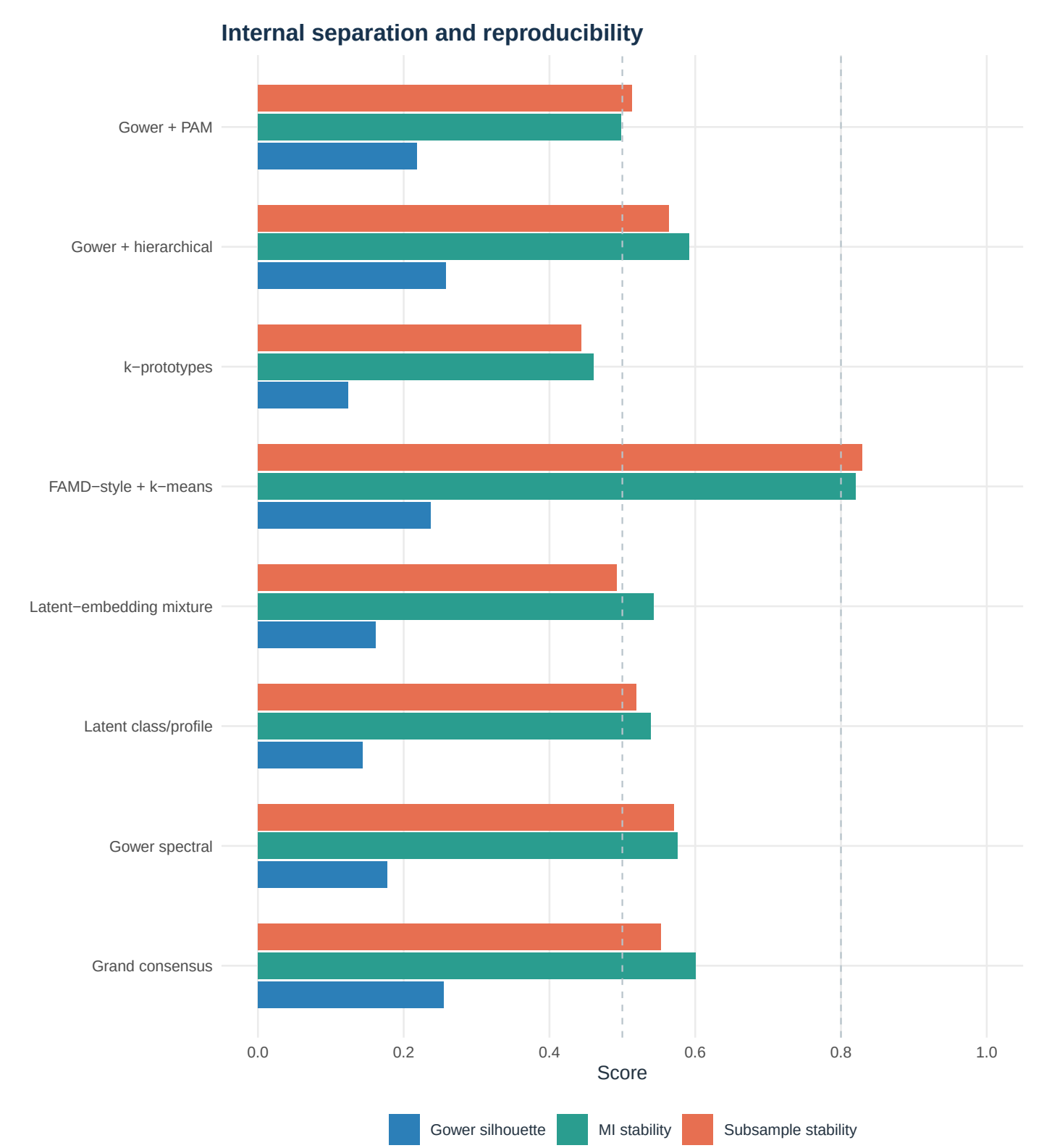
Method	ARI	NMI	Accuracy	Silhouette	MI stability	Subsample stability	Runtime (s)
Gower + PAM	NA	NA	NA%	0.218	0.498	0.513	0.00
Gower + hierarchical	NA	NA	NA%	0.258	0.592	0.563	0.00
k-prototypes	NA	NA	NA%	0.124	0.460	0.443	0.14
FAMD-style + k-means	NA	NA	NA%	0.237	0.820	0.829	0.00
Latent-embedding mixture	NA	NA	NA%	0.162	0.543	0.493	0.02
Latent class/profile	NA	NA	NA%	0.143	0.539	0.519	0.09
Gower spectral	NA	NA	NA%	0.177	0.576	0.570	0.00
Grand consensus	NA	NA	NA%	0.255	0.600	0.553	0.26

ARI, NMI, and accuracy are NA because no truth labels were supplied. This is expected for ordinary unsupervised analysis. Silhouette uses the first completed-data Gower distance. MI stability is mean pairwise ARI across imputation-specific partitions. Subsample stability compares refitted 80% subsets with the full-data solution on the same patients.

Runtime values are indicative on this machine and omit report rendering. Shared preprocessing makes exact cross-method timing comparisons approximate.

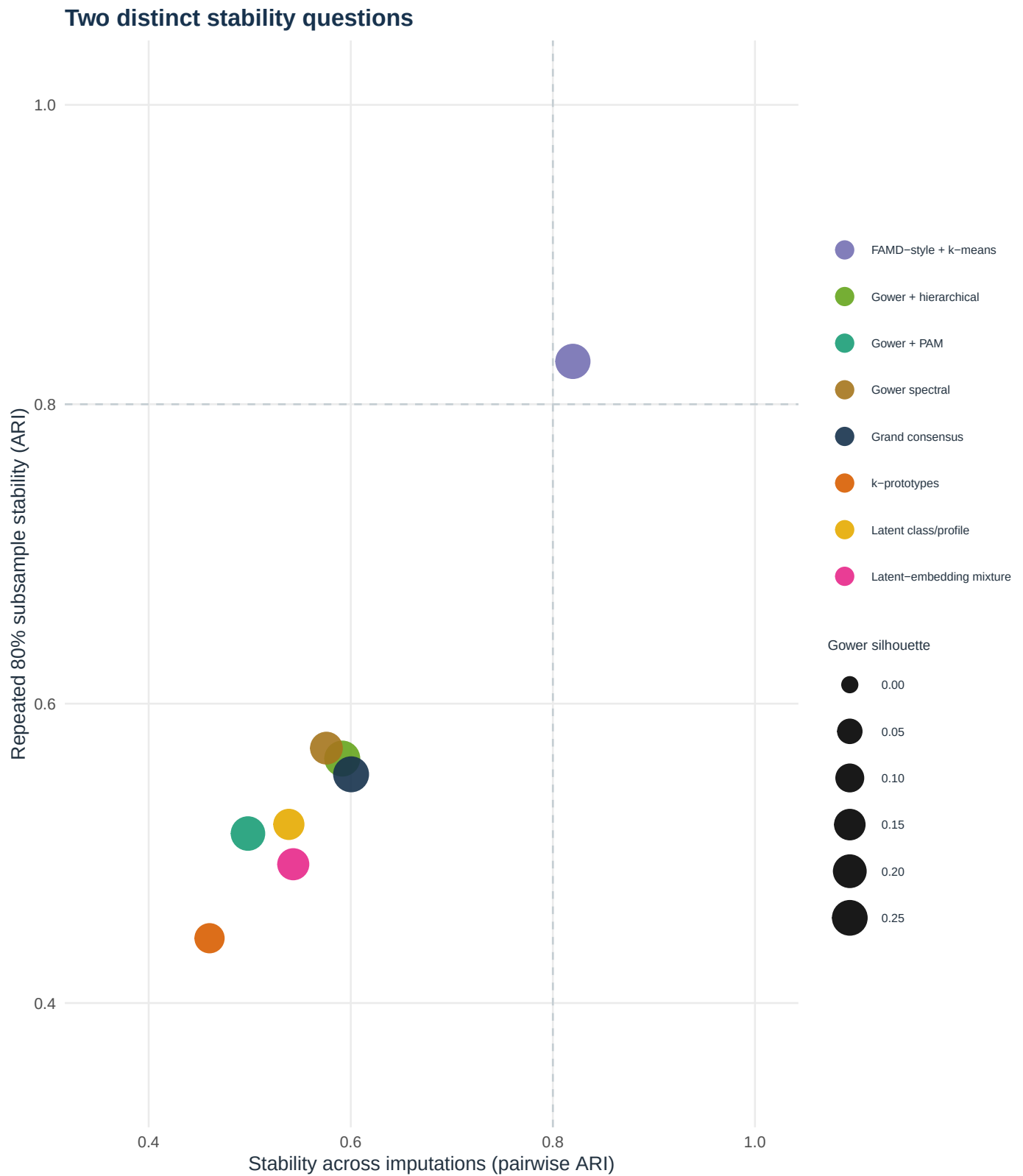
Internal performance and stability

No external truth labels were supplied



Stability and missing-data uncertainty

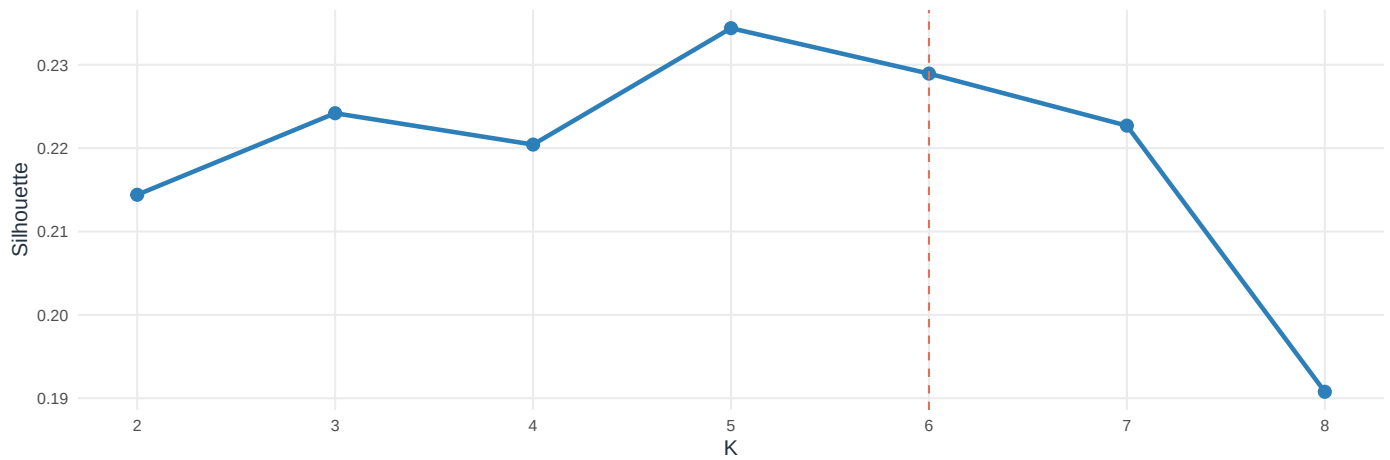
Strong recovery is useful only when assignments survive perturbation



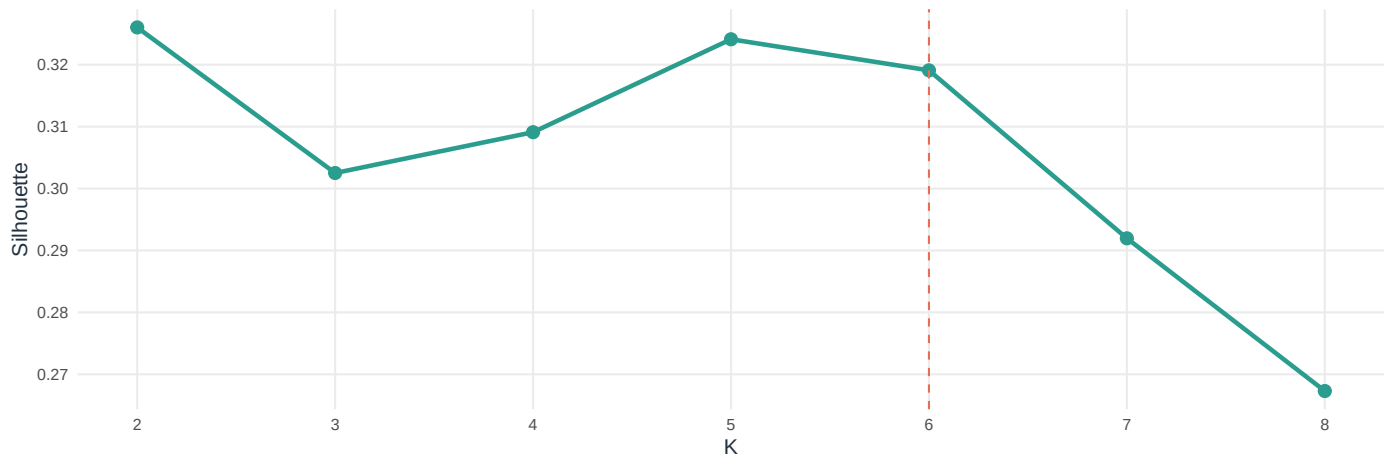
Choosing the number of clusters

No single criterion should determine K

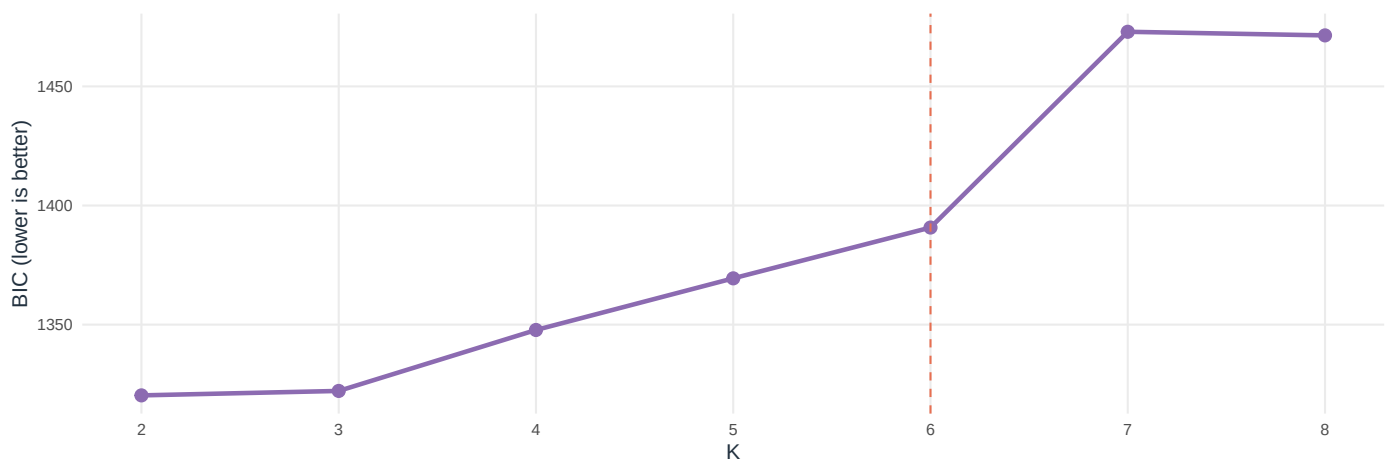
Gower + PAM



FAMD-style + k-means

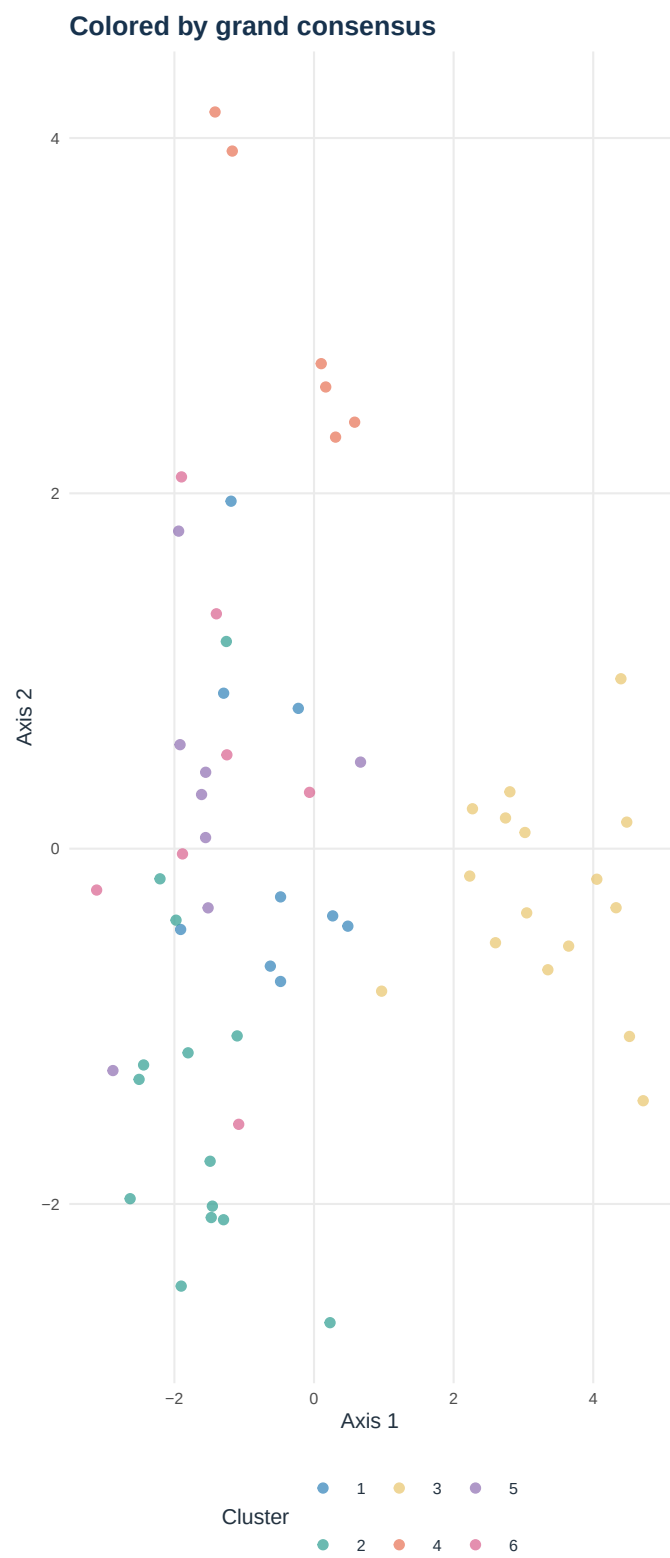
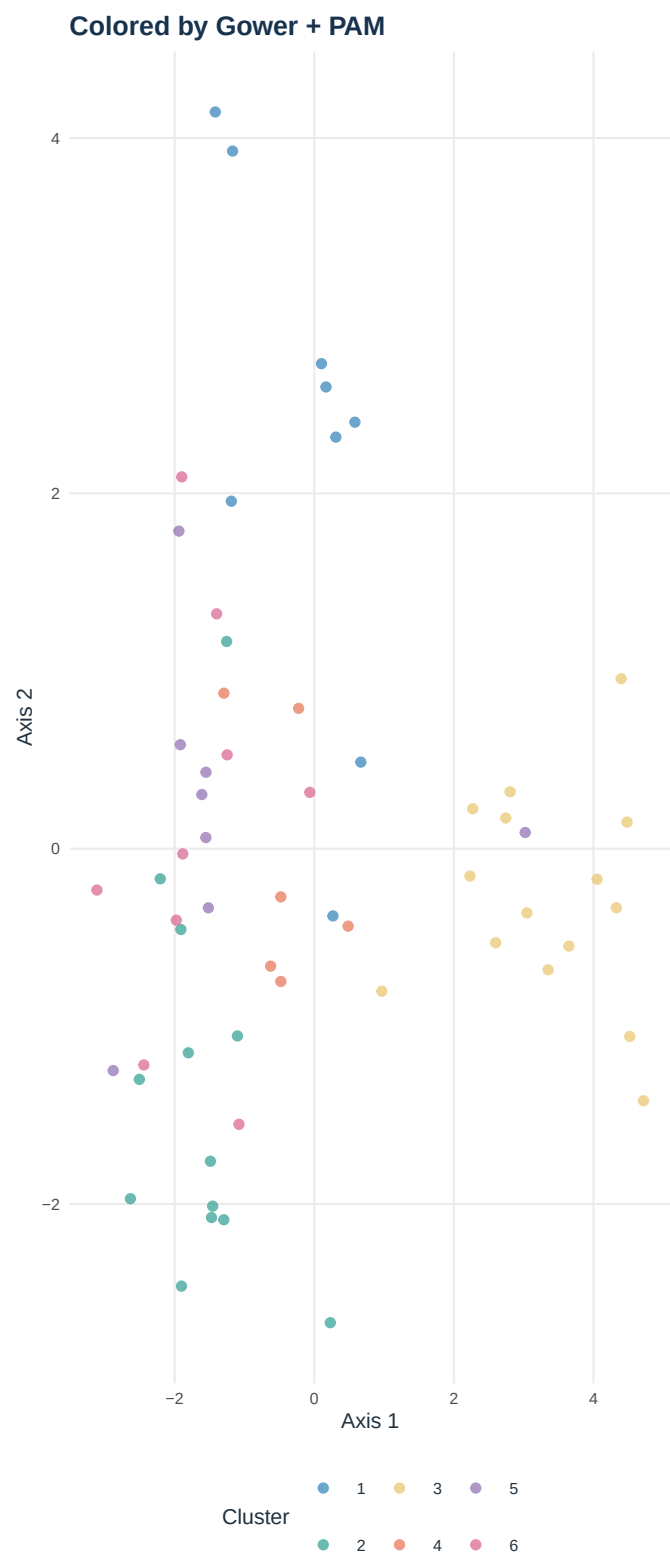


Latent mixture



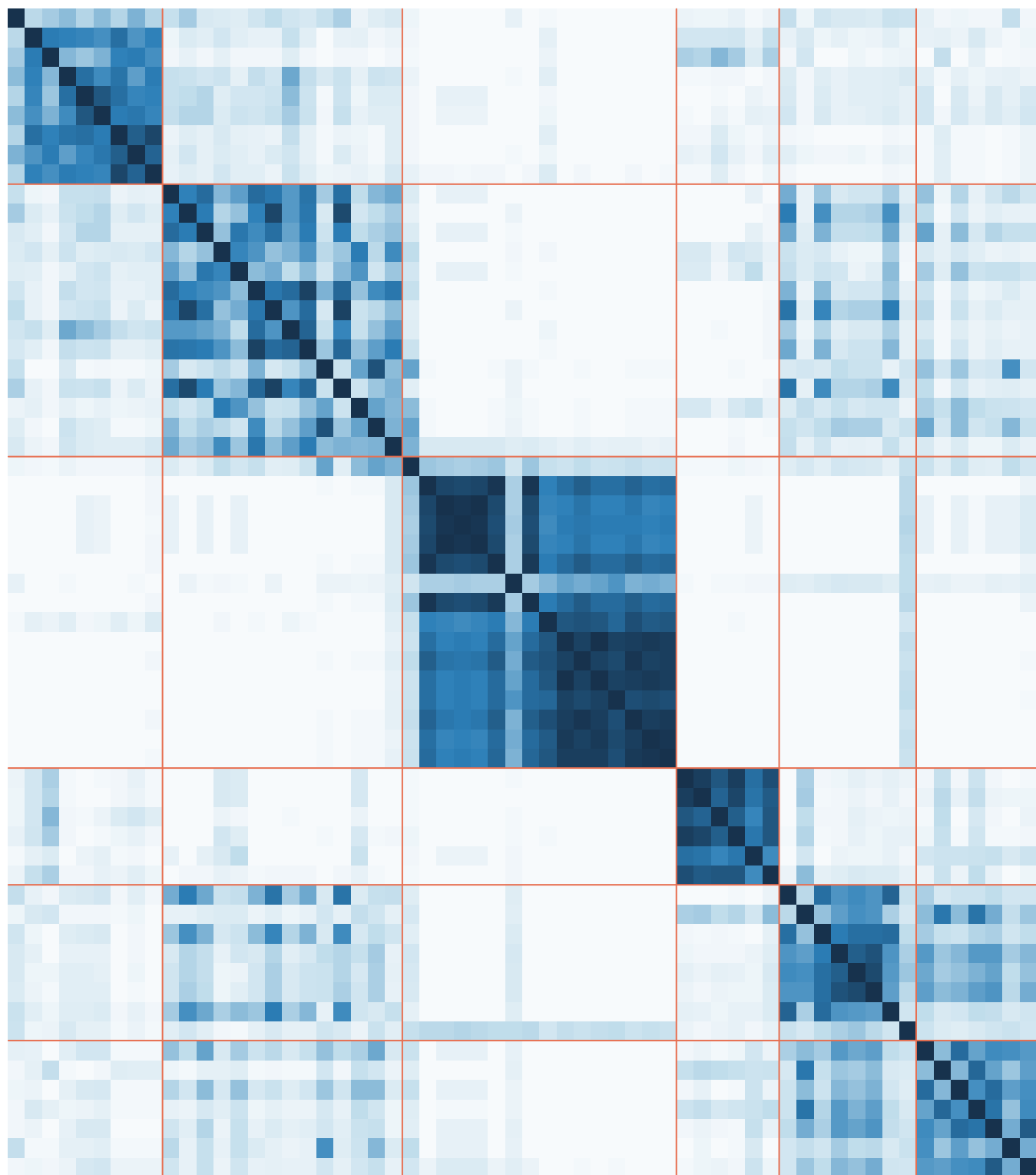
Mixed-data latent view

First two FAMD-style axes provide a common visual reference



Consensus co-assignment

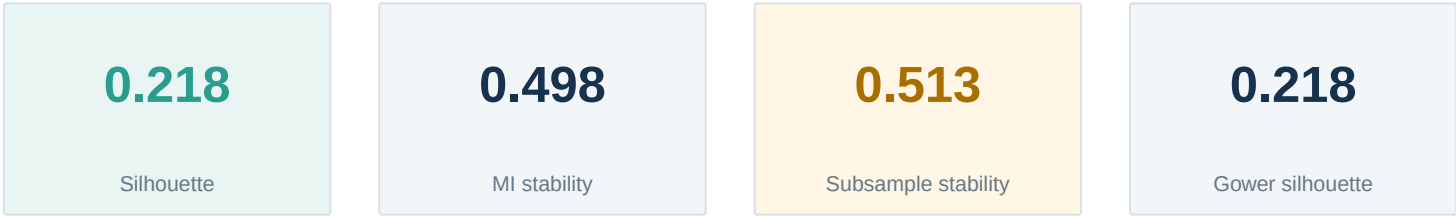
Cell color is the fraction of method-imputation partitions placing two patients together



Patients sorted by consensus cluster, then latent axis 1

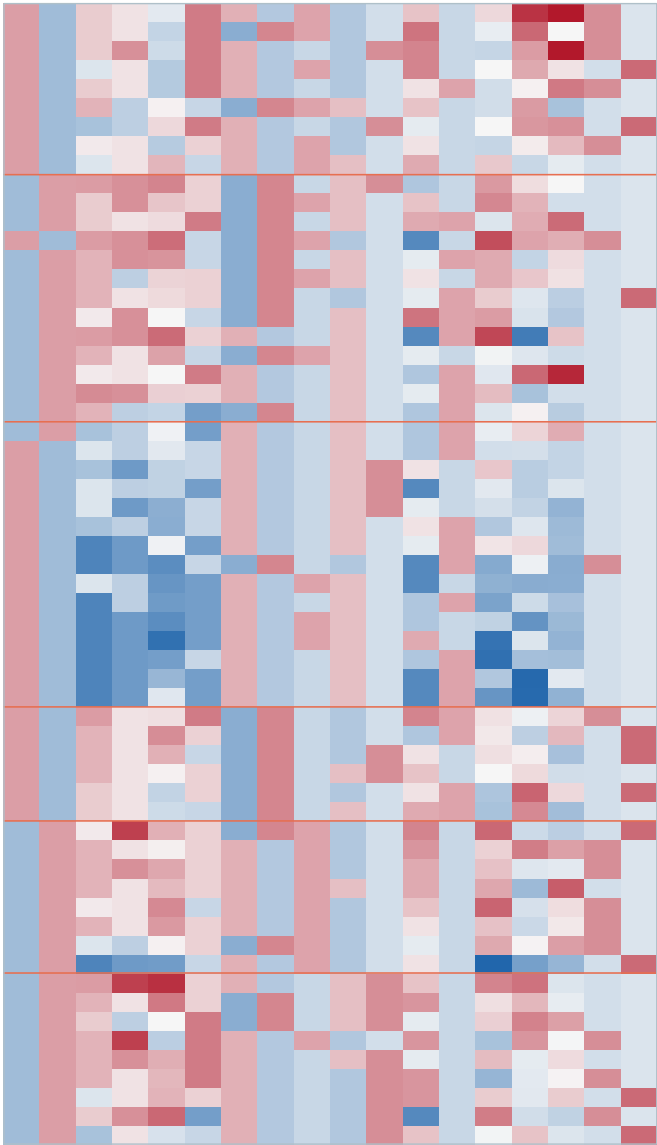
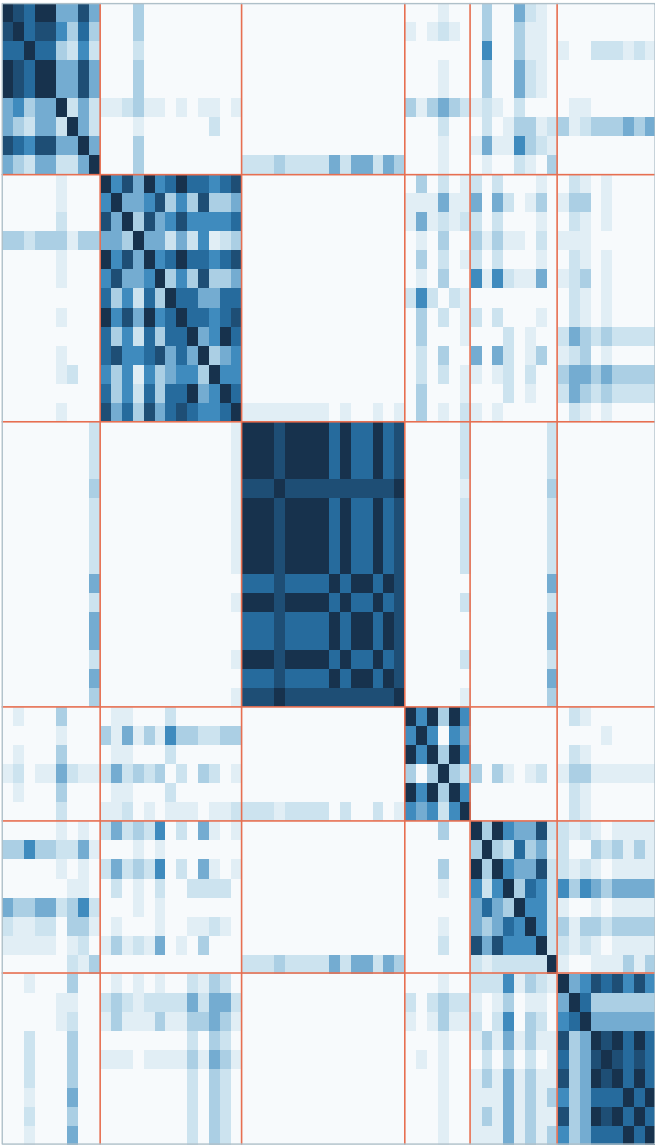
Method heatmaps: Gower + PAM

Left: stability across imputations. Right: patients by the most cluster-associated encoded features



Patient-patient co-assignment

Patient-feature heatmap (18 encoded columns)

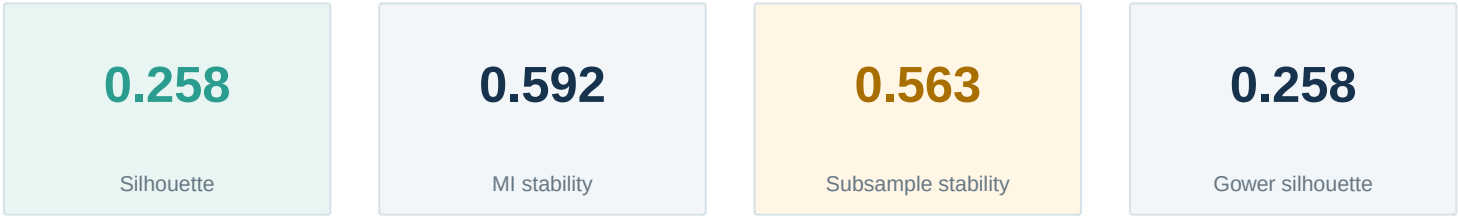


Cluster sizes: C1: 9 | C2: 13 | C3: 15 | C4: 6 | C5: 8 | C6: 9

Most associated encoded columns: diabetes=No, diabetes=Yes, medication_count, functional_limitation_l1_l5, bmi_kg_m2, symptom_severity_l1_l4, biomarker_positive=No, biomarker_positive=Yes

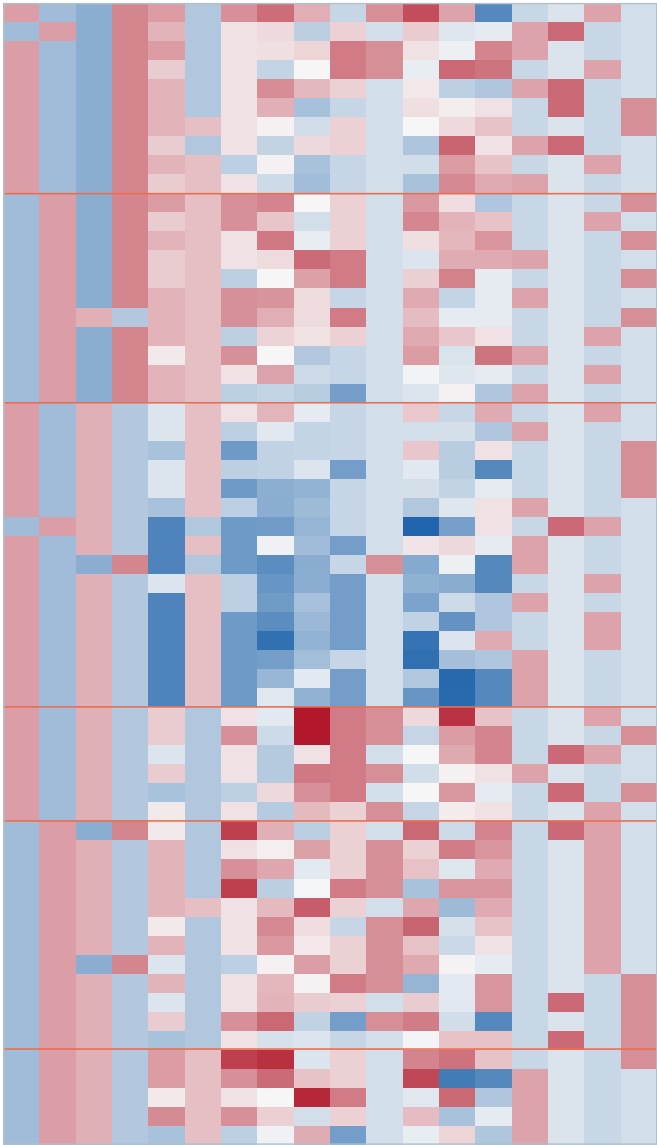
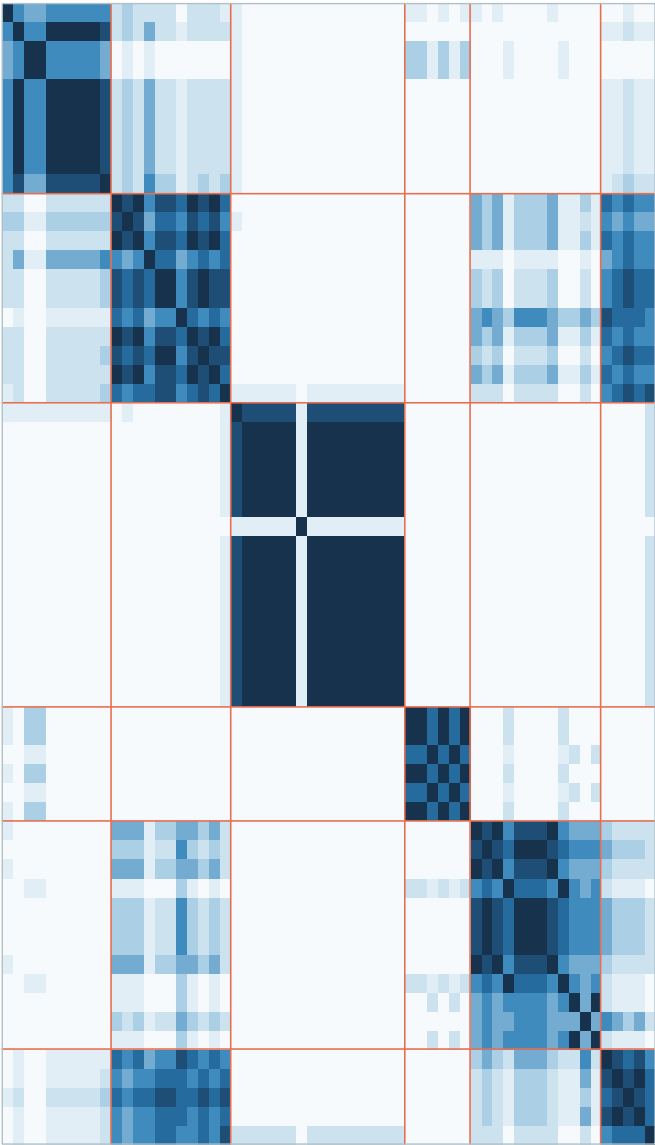
Method heatmaps: Gower + hierarchical

Left: stability across imputations. Right: patients by the most cluster-associated encoded features



Patient-patient co-assignment

Patient-feature heatmap (18 encoded columns)

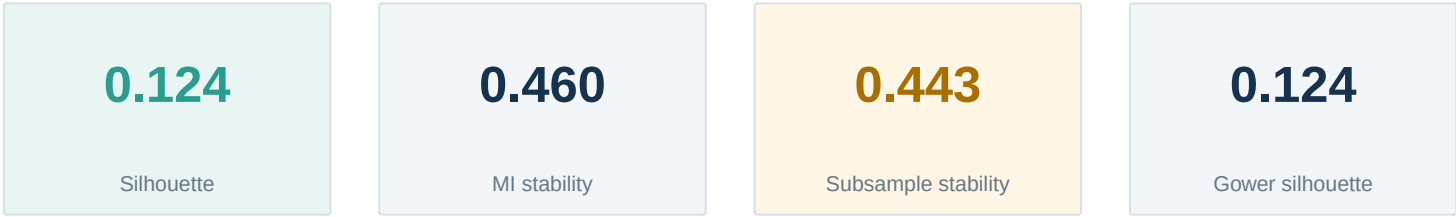


Cluster sizes: C1: 10 | C2: 11 | C3: 16 | C4: 6 | C5: 12 | C6: 5

Most associated encoded columns: diabetes=No, diabetes=Yes, biomarker_positive=No, biomarker_positive=Yes, medication_count, care_setting=outpatient, functional_limitation_l1_l5, bmi_kg_m2

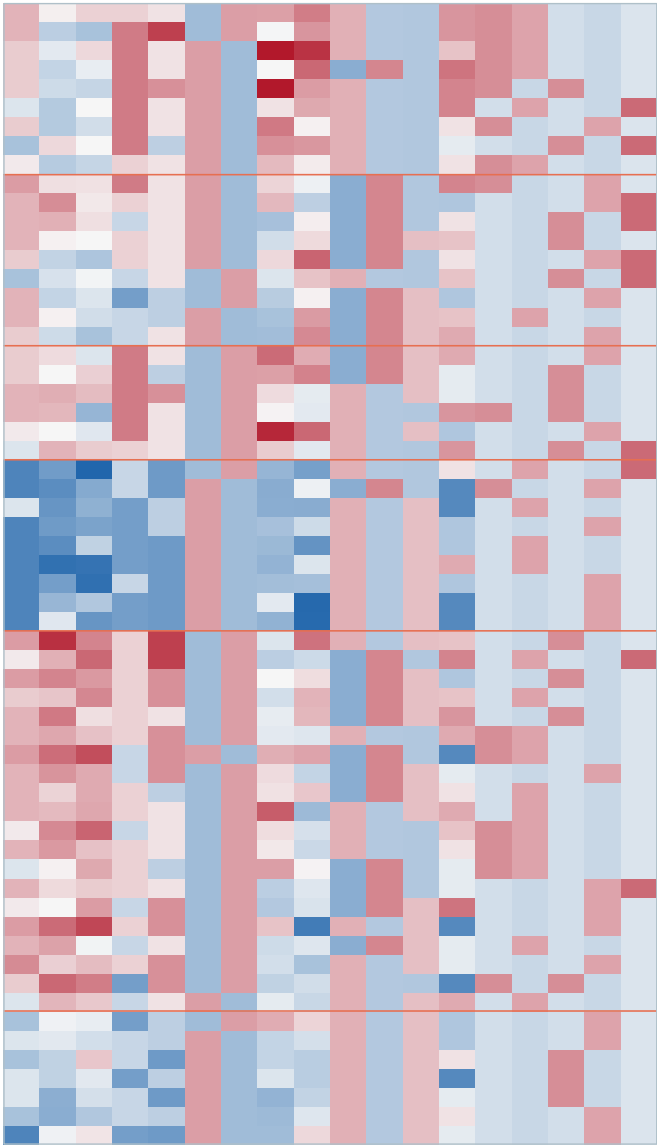
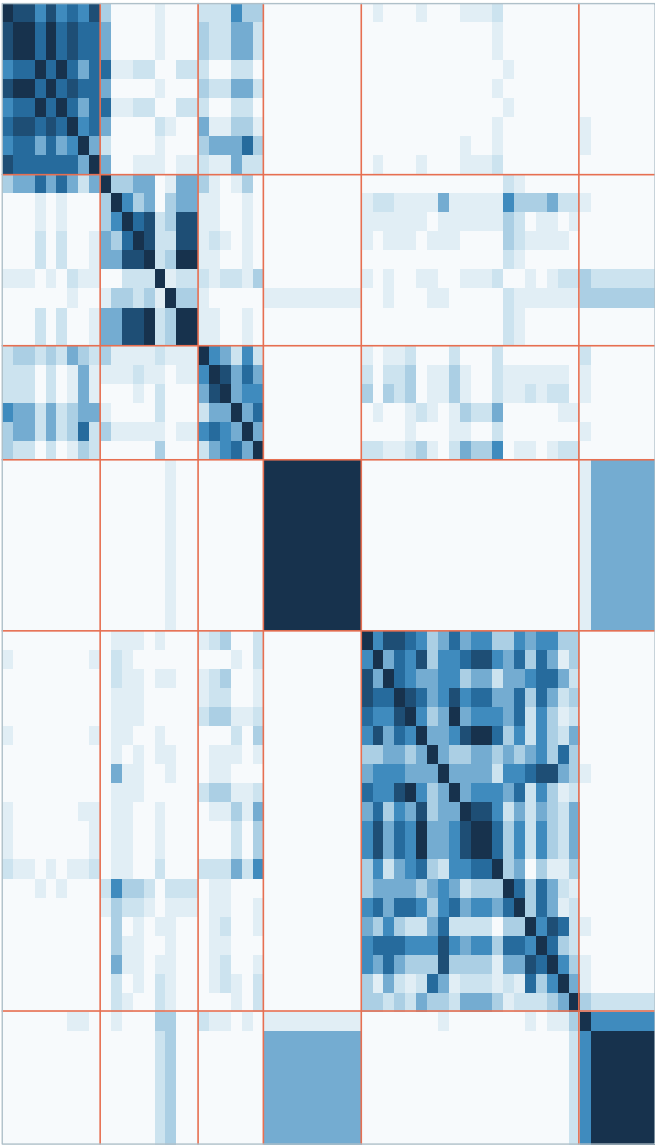
Method heatmaps: k-prototypes

Left: stability across imputations. Right: patients by the most cluster-associated encoded features



Patient-patient co-assignment

Patient-feature heatmap (18 encoded columns)

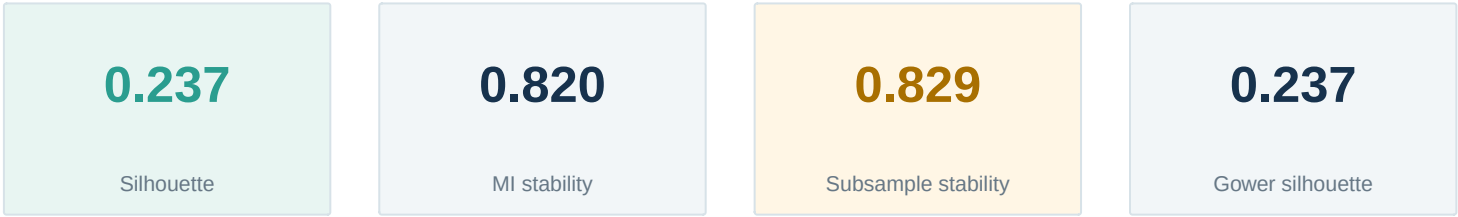


Cluster sizes: C1: 9 | C2: 9 | C3: 6 | C4: 9 | C5: 20 | C6: 7

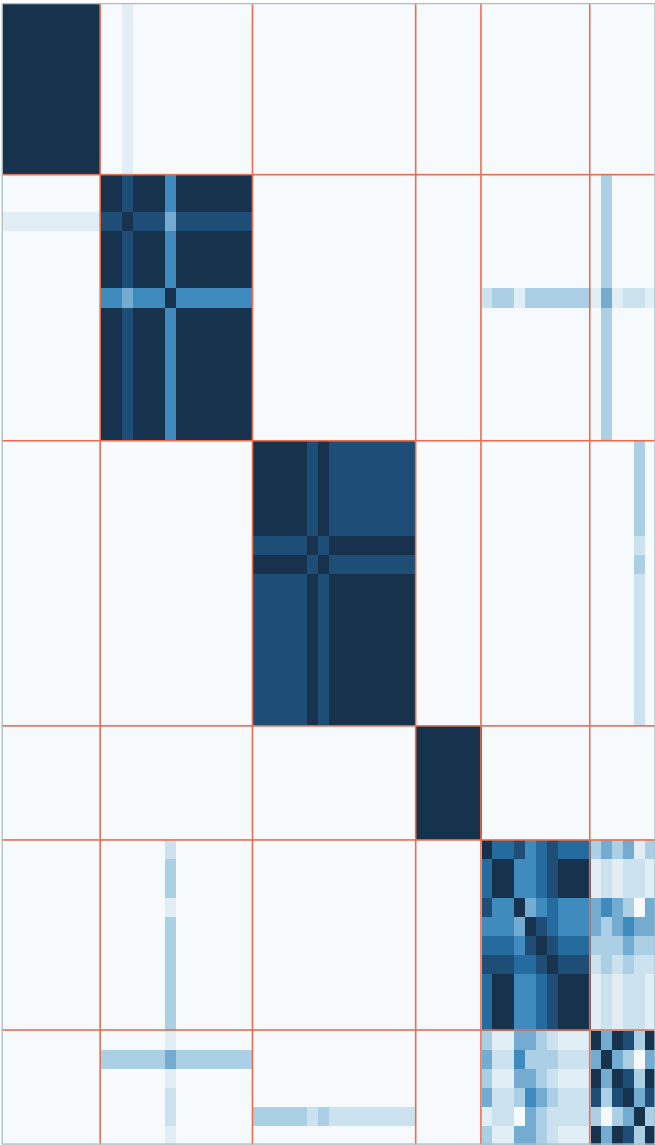
Most associated encoded columns: medication_count, bmi_kg_m2, systolic_bp_mmhg, symptom_severity_l1_l4, functional_limitation_l1_l5, diabetes=No, diabetes=Yes, crp_mg_l

Method heatmaps: FAMD-style + k-means

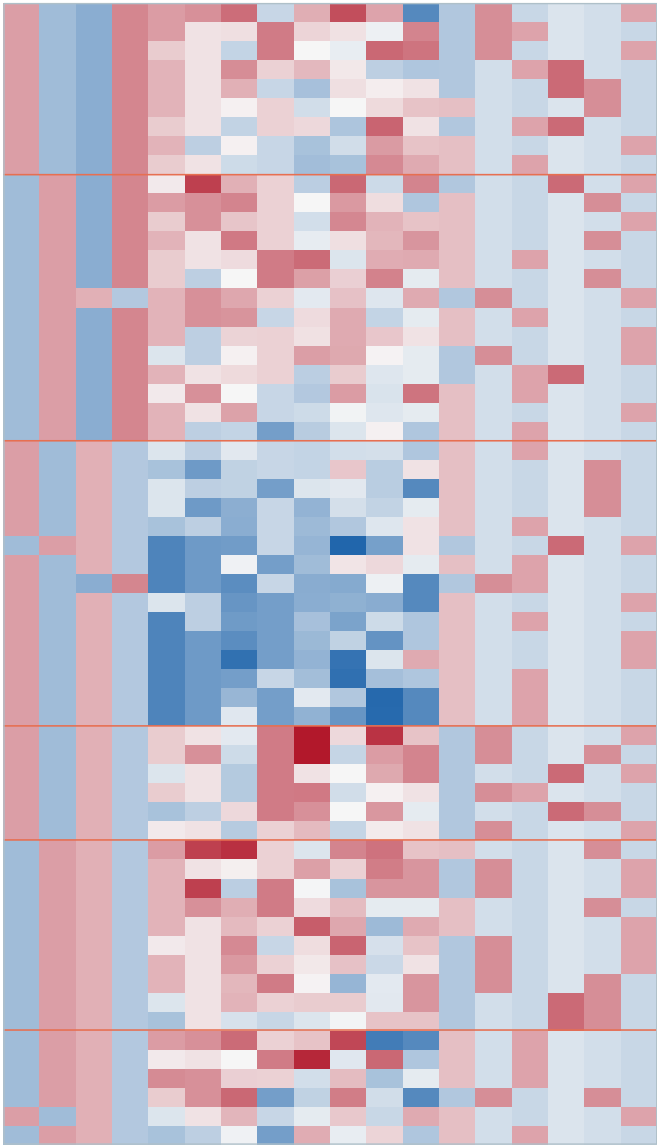
Left: stability across imputations. Right: patients by the most cluster-associated encoded features



Patient-patient co-assignment



Patient-feature heatmap (18 encoded columns)

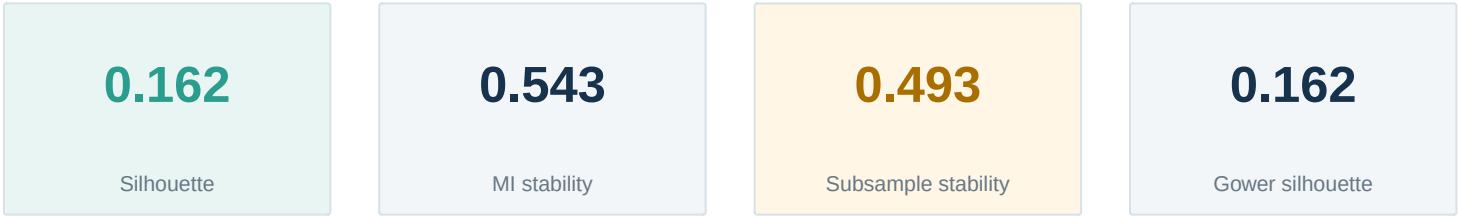


Cluster sizes: C1: 9 | C2: 14 | C3: 15 | C4: 6 | C5: 10 | C6: 6

Most associated encoded columns: diabetes=No, diabetes=Yes, biomarker_positive=No, biomarker_positive=Yes, medication_count, functional_limitation_l1_l5, bmi_kg_m2, symptom_severity_l1_l4

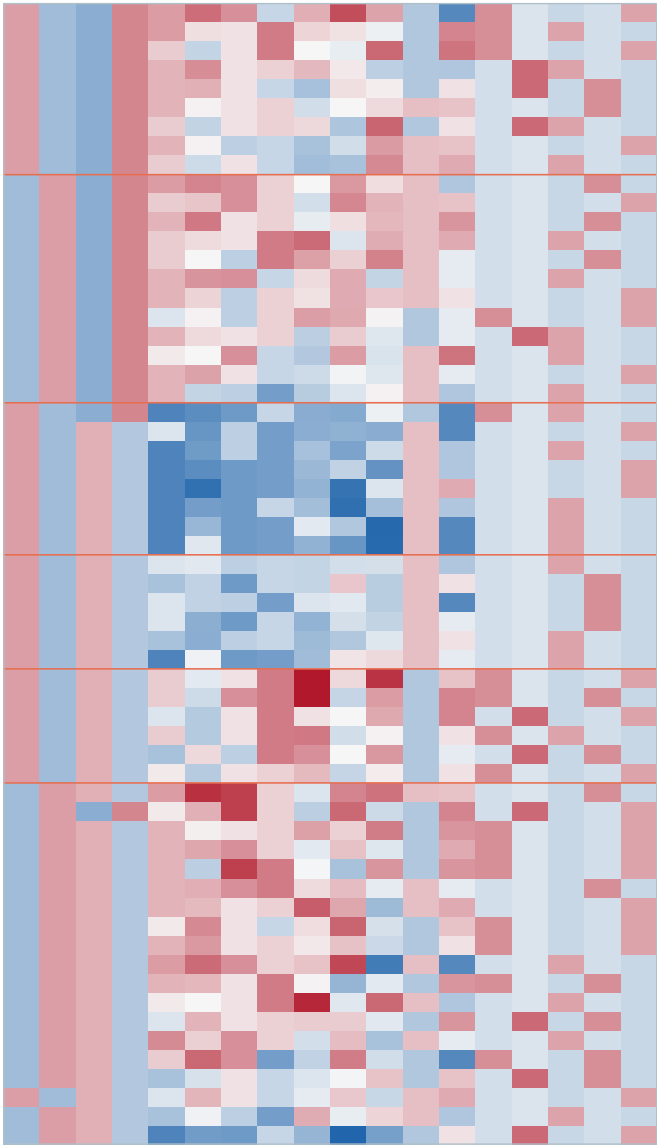
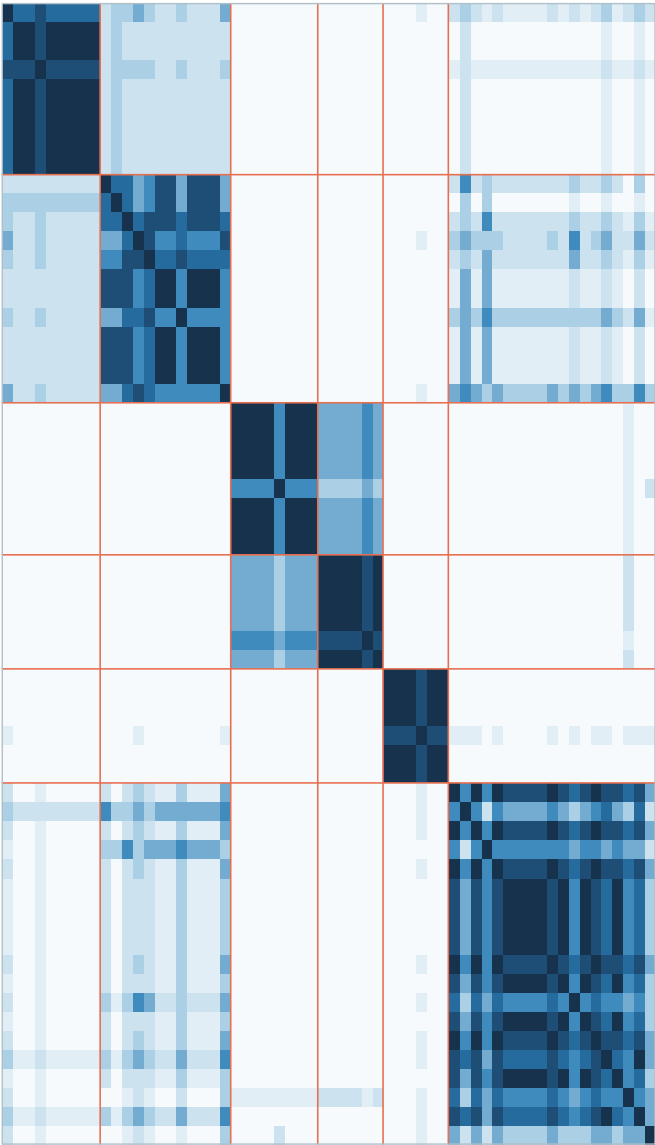
Method heatmaps: Latent–embedding mixture

Left: stability across imputations. Right: patients by the most cluster–associated encoded features



Patient–patient co–assignment

Patient–feature heatmap (18 encoded columns)

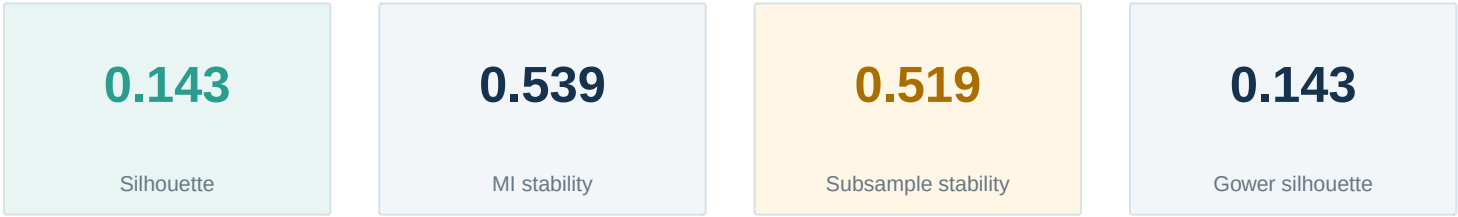


Cluster sizes: C1: 9 | C2: 12 | C3: 8 | C4: 6 | C5: 6 | C6: 19

Most associated encoded columns: diabetes=No, diabetes=Yes, biomarker_positive=No, biomarker_positive=Yes, medication_count, bmi_kg_m2, functional_limitation_I1_I5, symptom_severity_I1_I4

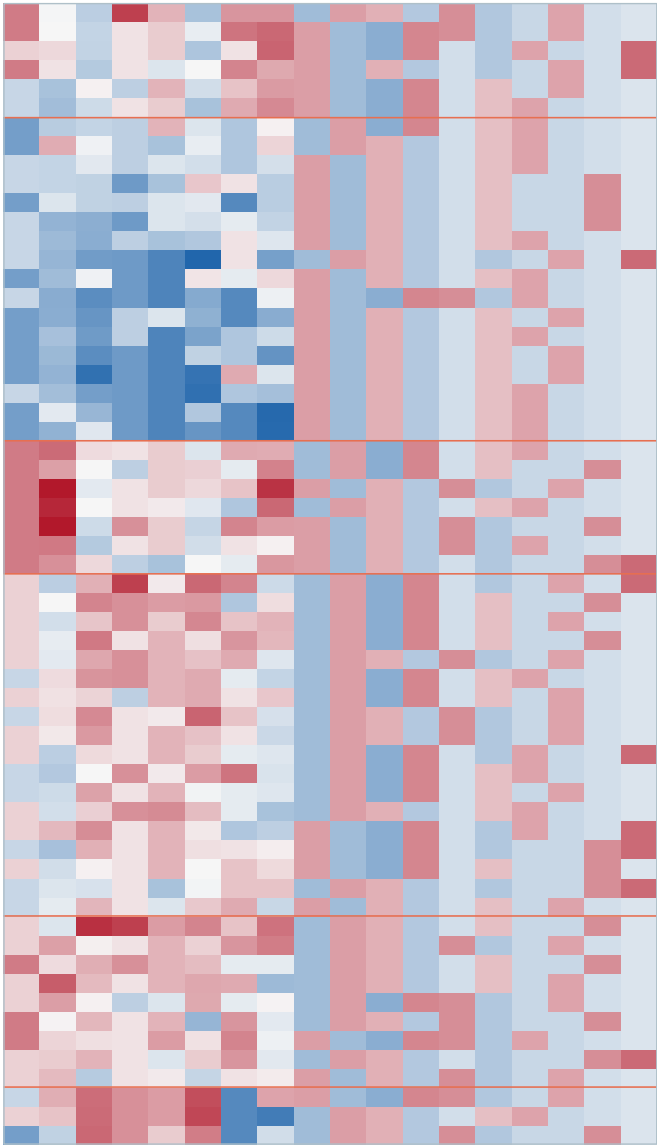
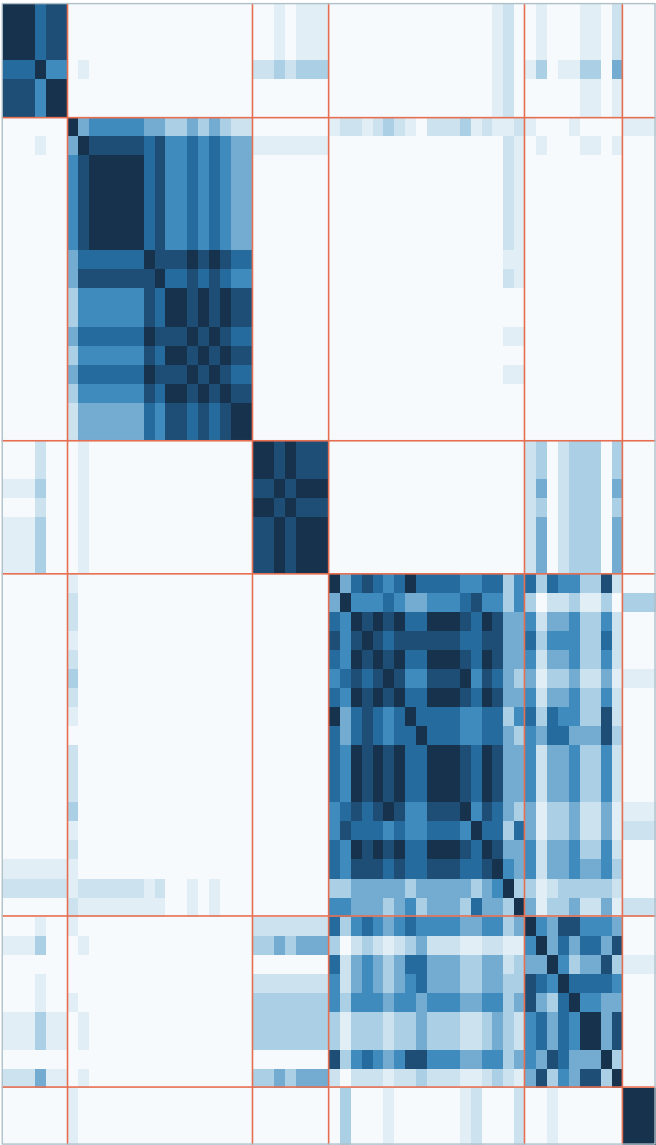
Method heatmaps: Latent class/profile

Left: stability across imputations. Right: patients by the most cluster-associated encoded features



Patient-patient co-assignment

Patient-feature heatmap (18 encoded columns)

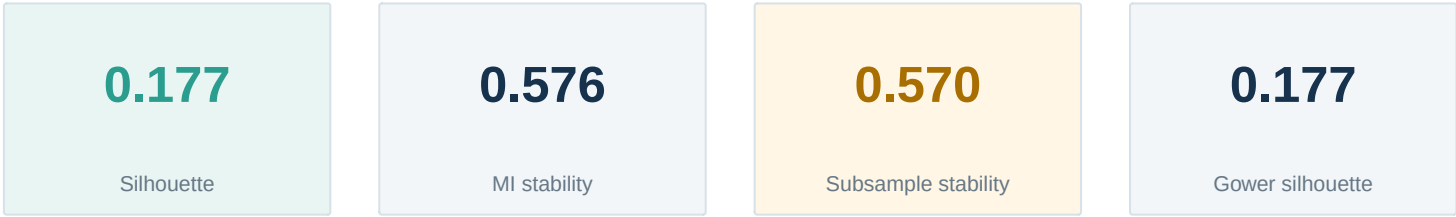


Cluster sizes: C1: 6 | C2: 17 | C3: 7 | C4: 18 | C5: 9 | C6: 3

Most associated encoded columns: symptom_severity_l1_l4, crp_mg_l, bmi_kg_m2, functional_limitation_l1_l5, medication_count, systolic_bp_mmhg, visits_past_year, age_years

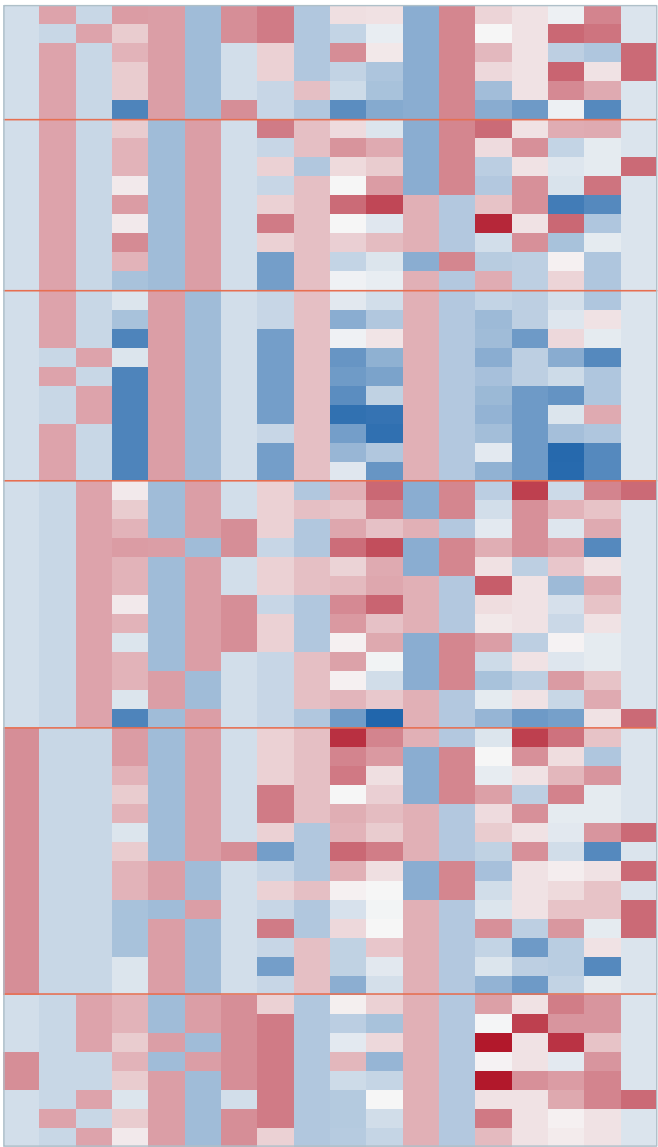
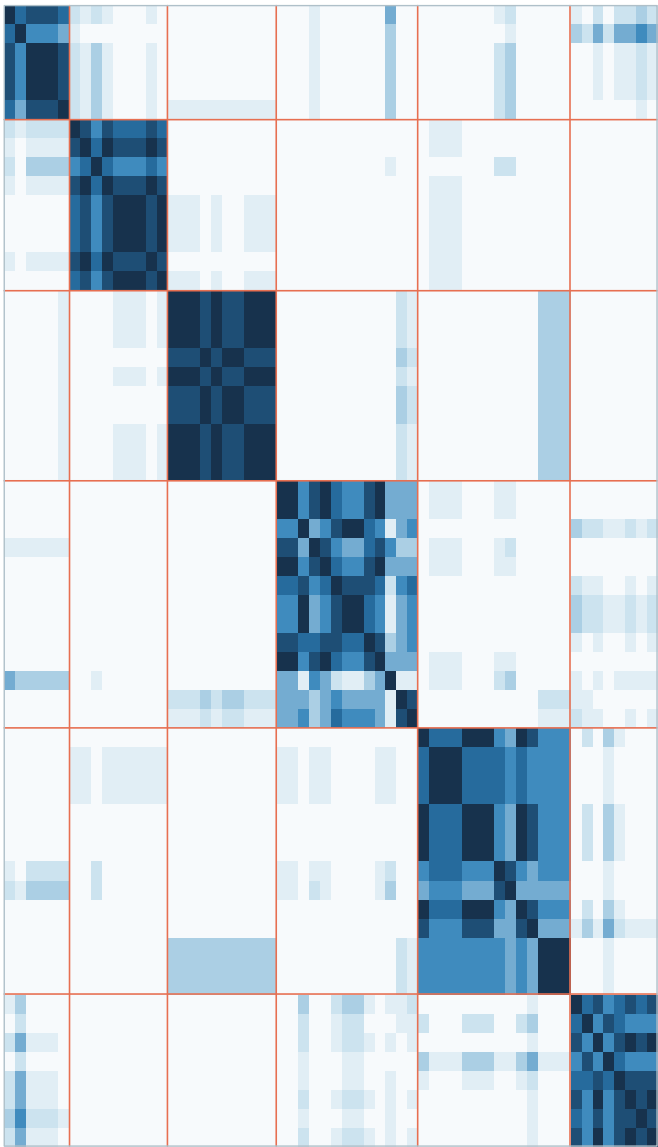
Method heatmaps: Gower spectral

Left: stability across imputations. Right: patients by the most cluster-associated encoded features



Patient-patient co-assignment

Patient-feature heatmap (18 encoded columns)

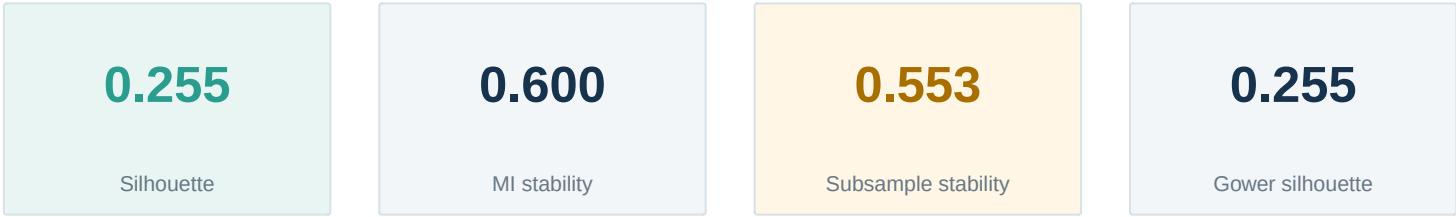


Cluster sizes: C1: 6 | C2: 9 | C3: 10 | C4: 13 | C5: 14 | C6: 8

Most associated encoded columns: smoking_status=current, smoking_status=never, smoking_status=former, medication_count, diabetes=No, diabetes=Yes, care_setting=urgent, symptom_severity_l1_l4

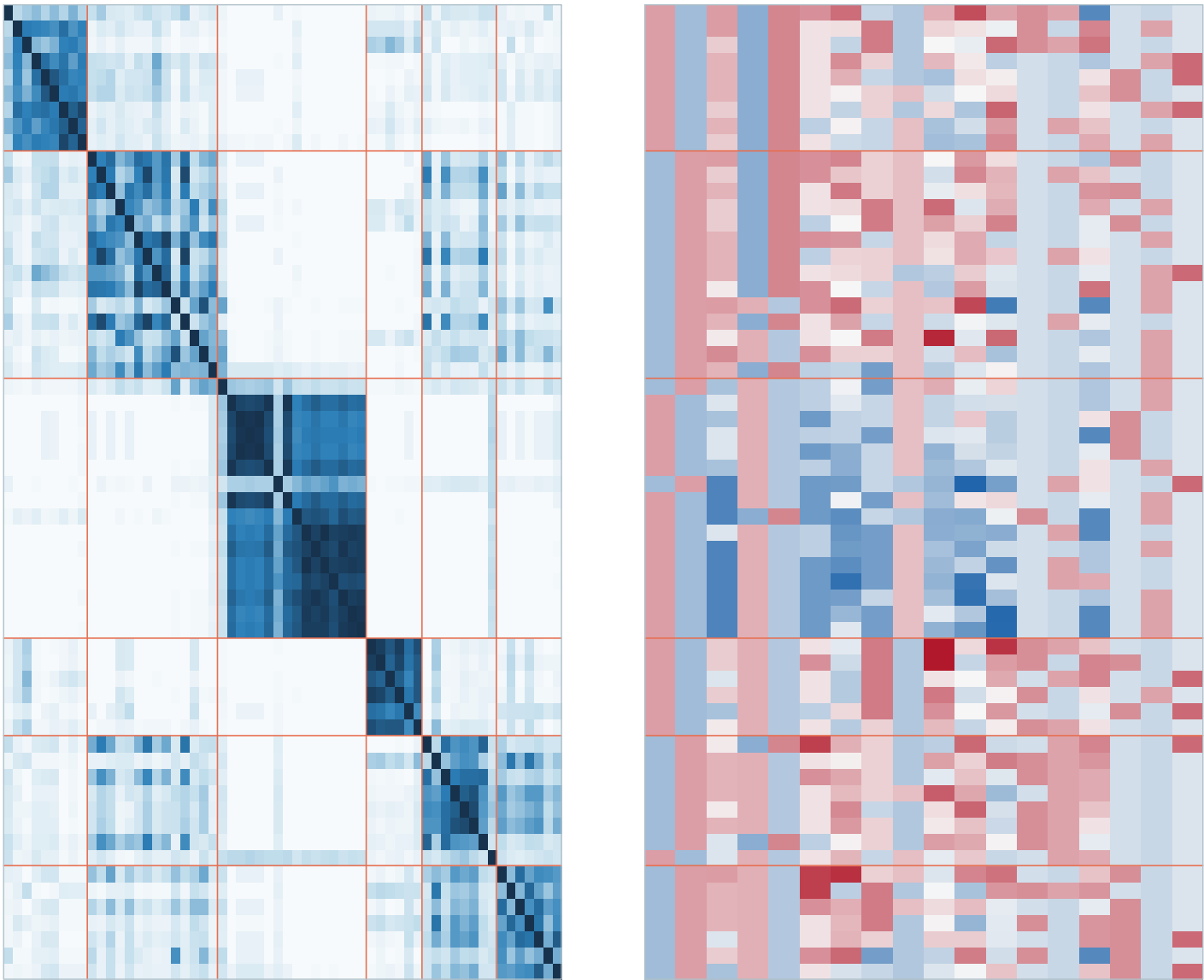
Method heatmaps: Grand consensus

Left: stability across imputations. Right: patients by the most cluster-associated encoded features



Patient-patient co-assignment

Patient-feature heatmap (18 encoded columns)

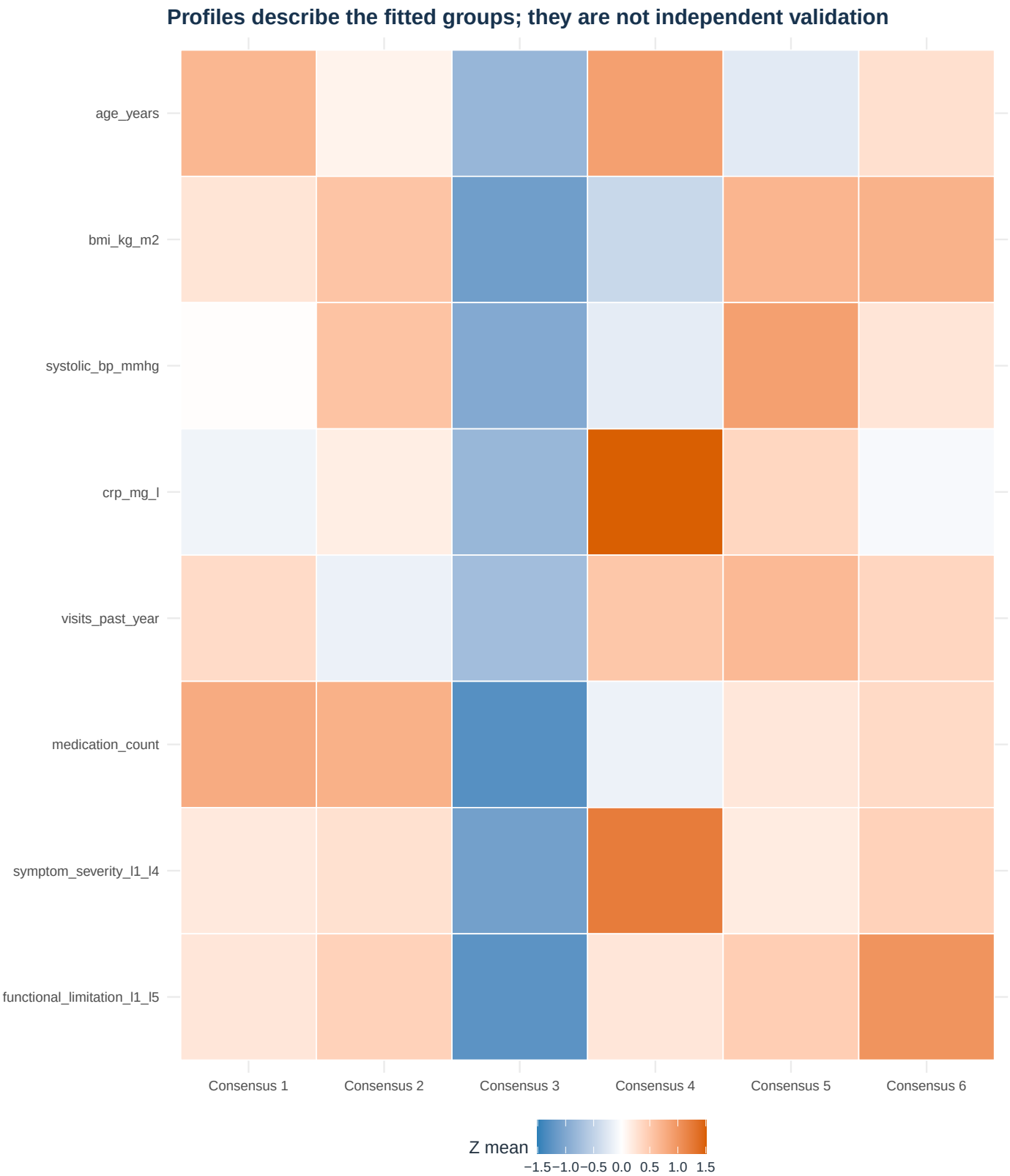


Cluster sizes: C1: 9 | C2: 14 | C3: 16 | C4: 6 | C5: 8 | C6: 7

Most associated encoded columns: diabetes=No, diabetes=Yes, medication_count, biomarker_positive=No, biomarker_positive=Yes, functional_limitation_l1_l5, bmi_kg_m2, symptom_severity_l1_l4

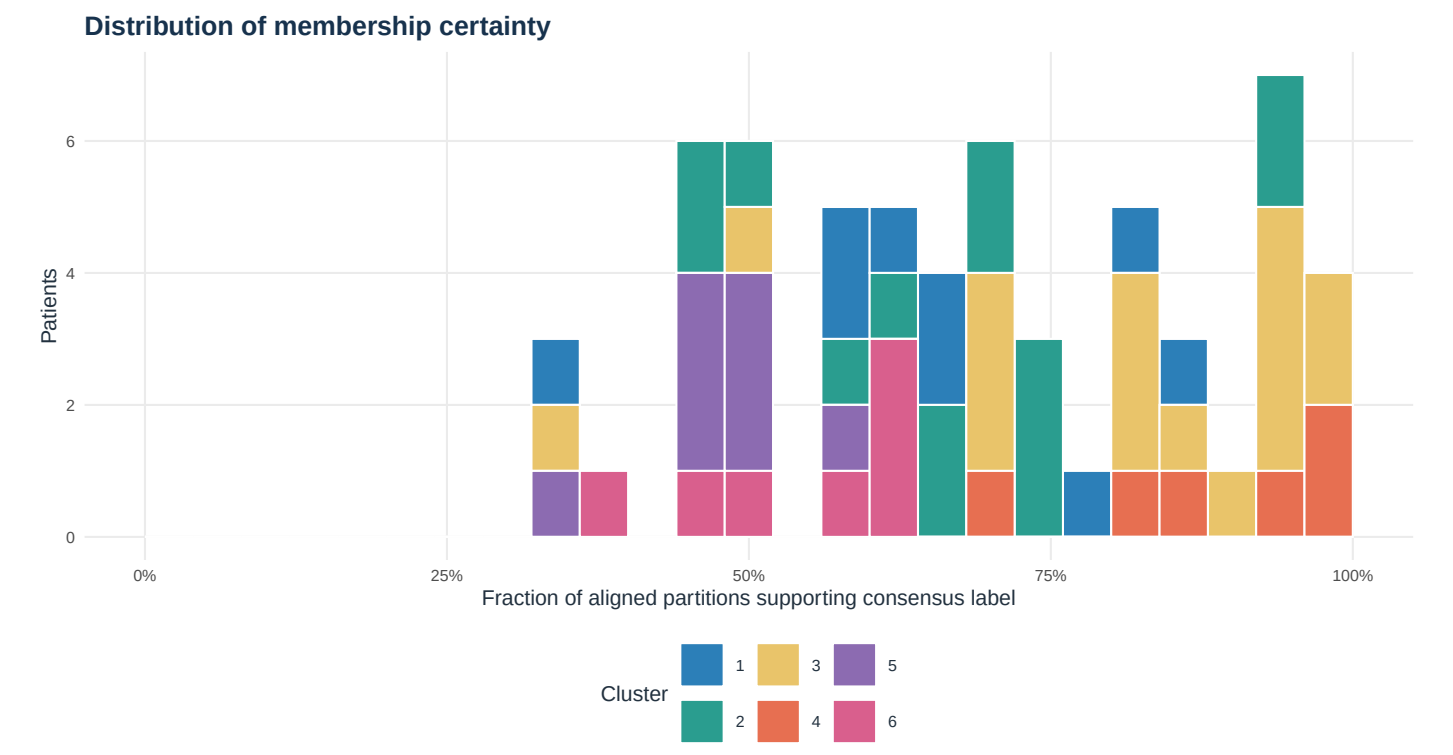
Consensus cluster profiles

Mean standardized values for selected numeric or ordinal features



Membership uncertainty

Consensus support exposes patients whose assignments are not robust

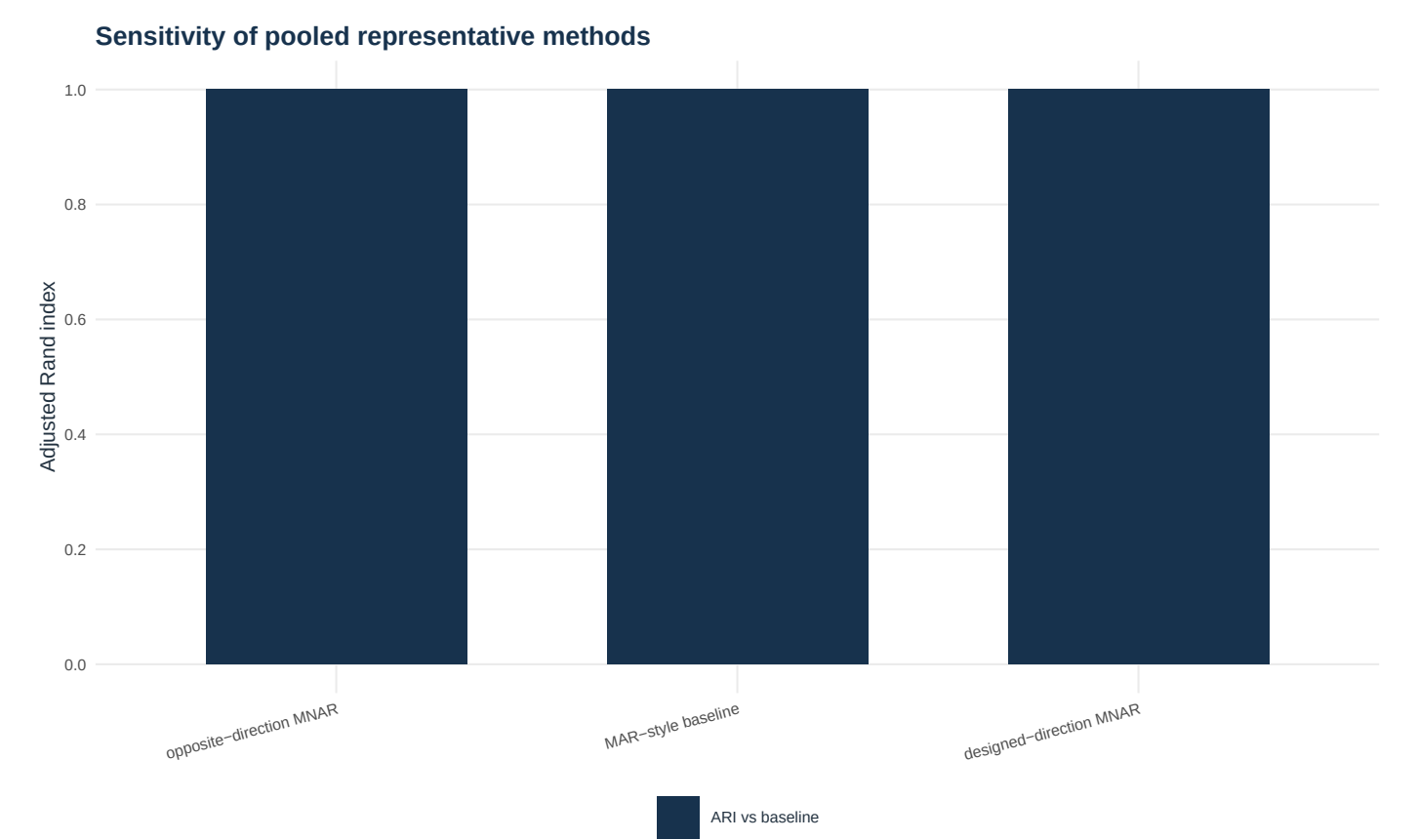


Ten least-certain patients

Patient	Consensus	Certainty
P027	3	32.1%
P043	5	32.1%
P022	1	33.9%
P030	6	39.3%
P021	5	44.6%
P024	5	44.6%
P040	2	44.6%
P047	6	44.6%
P028	5	46.4%
P054	2	46.4%

MNAR sensitivity analysis

Only originally missing values in 0 dictionary-designated features are shifted



Delta	Scenario	ARI vs baseline	Reassigned
-1	opposite-direction MNAR	1.000	0.0%
+0	MAR-style baseline	1.000	0.0%
+1	designed-direction MNAR	1.000	0.0%

No `mnar_direction` values were specified, so the three scenarios are identical. Edit `feature_dictionary_used.csv` to mark plausible directions as `-1` or `+1`, then rerun the CSV command to obtain an informative sensitivity analysis.

How to interpret this benchmark

What transfers to a real cohort and what does not

- External recovery scores transfer only as evidence that the implementations work on this simulation. A real cohort has no hidden truth labels, and good silhouette values do not establish biological subtypes.
- The stochastic nearest-neighbor hot-deck procedure preserves mixed-data values and propagates some imputation uncertainty. For publication-grade analysis, consider a carefully specified MICE or joint-model imputation and repeat the same consensus workflow.
- The native latent class/profile model uses diagonal Gaussian distributions for continuous/log-count variables and multinomials for ordinal and nominal variables. It is intentionally parsimonious; specialized count and ordinal likelihoods may be preferable.
- The FAMD-style implementation is transparent and dependency-light but is not a drop-in reproduction of every weighting convention used by FactoMineR or PCAmixdata.
- Cluster-feature differences are descriptive because those features constructed the clusters. Validate with prespecified variables not used for clustering, then replicate in an external cohort.
- If the scientific goal is prediction, treatment selection, or outcome risk, a supervised model may be more appropriate than clustering. Clusters should not be interpreted causally.

Rerunning with your own data

The project is designed to be edited rather than treated as a black box

Fastest complete rerun

```
Rscript analyze_csv.R --input path/to/patients.csv --k auto
```

- Put one patient per row. Use `--id-column` for an identifier, `--truth-column` only for optional external labels, and `--exclude` for outcomes, site, batch, or other columns that should not define clusters.
- A dictionary is inferred when `--dictionary` is omitted and saved as `feature_dictionary_used.csv`. Review it, especially numeric ordinal variables, then rerun with `--dictionary` pointing to the edited file.
- Allowed types are continuous, discrete, ordinal, categorical, and binary. The domain column controls block weighting. Optional role and mnar_direction columns support reporting and sensitivity analysis.
- Use `--k auto` for a rank aggregation of two silhouette diagnostics and latent-mixture BIC, or provide an integer. Always inspect K-selection, stability, cluster sizes, uncertainty, and clinical coherence together.
- Increase `m_imputations` and `n_subsamples` for a final analysis. Eight and ten are intentionally modest defaults for an executable demonstration.
- Machine-readable outputs include method metrics, cluster assignments, membership certainty, missingness, K selection, MNAR sensitivity, resampling stability, the dictionary used, and a run manifest.

Project output inventory

Files produced by the reproducible pipeline

File	Purpose
feature_dictionary_used.csv	Feature types and domains actually used
auto_k_diagnostics.csv (if --k auto)	Diagnostics used when --k auto is selected
run_manifest.csv	Input, output, K, ID, and truth settings
results/method_metrics.csv	External, internal, stability, and runtime metrics
results/cluster_assignments.csv	Consensus and method-specific patient labels
results/membership_certainty.csv	Patient-level assignment support
results/k_selection.csv	Candidate-K diagnostics
results/mnar_sensitivity.csv	Directional departure-from-MAR scenarios
results/analysis_results.rds	Complete report-ready analysis object
clustering_report.pdf	This rendered benchmark report

All run-specific files are written below the chosen --output-dir, except when --pdf explicitly points elsewhere.

Reproducibility record

Software information captured during execution

R version 4.6.1 (2026-06-24)
Platform: aarch64-apple-darwin23
Running under: macOS Tahoe 26.6.2

Matrix products: default
BLAS: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1

locale:
[1] C.UTF-8/C.UTF-8/C.UTF-8/C.UTF-8/C.UTF-8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats graphics grDevices utils datasets methods base

loaded via a namespace (and not attached):
[1] compiler_4.6.1 cluster_2.1.8.2

Analysis timestamp (UTC): 2026-09-03 02:02:52